CLOUD COMPUTING & IoT CLOUD TECHNOLOGY

40 Important Questions - 7-Mark A+ Exam Answer Style

Numbered points | Short explanations | Examples | Conclusions | Exam suggestions

Source Basis: Prepared strictly from the uploaded 103-page lecture PDF (w11.pdf). Historical dates, rankings and product descriptions are retained as lecture content and are not presented as current market facts.

Exam-ready revision handbook

How to Write a Perfect 7-Mark Answer

1. **Definition/Introduction (1 mark):** Start with two direct lines defining the concept.
2. **Numbered technical points (4-5 marks):** Write 6-7 precise points. Bold the keyword, then add one short explanation.
3. **Example/diagram/table (1 mark):** Use a source-based example or a small labeled diagram.
4. **Conclusion (0.5 mark):** End with the main result or importance of the topic.
5. **Presentation booster:** Underline technical terms, leave spacing, and use a comparison table when the question says compare.

A+ Rule: Do not write one long paragraph. Use numbered points, correct full forms, one source-based example and a two-line conclusion.

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Q1. Trace the evolution of cloud computing.

Source basis: uploaded lecture PDF, pages 2-4 | Suggested answer value: 7 marks

Introduction

Cloud computing evolved gradually from shared mainframes, networking and virtualization into Internet-based, on-demand computing services.

Main Points

1. **Time-shared mainframes (1950s):** Many users shared the processing capacity of a powerful central computer.
2. **ARPANET (1969):** Wide-area networking created the foundation for remote access to computing resources.
3. **Virtual machines (1970s):** IBM virtualization allowed one physical machine to behave like several independent machines.
4. **Internet and VPN expansion (1990s):** Global connectivity and virtual private networks enabled remote, secure resource access.
5. **Term cloud computing (1996-97):** The expression began to describe computing resources delivered through networks.
6. **Commercial cloud services (1999-2006):** Salesforce delivered applications online; AWS and Amazon EC2 introduced web services and rentable compute capacity.
7. **Modern platforms (2008 onward):** Google App Engine and Microsoft Azure expanded platform and application hosting in the cloud.

Example

A company can rent an Amazon EC2 virtual server instead of purchasing and maintaining a physical server.

Conclusion

Thus, cloud computing is the result of convergence among networking, virtualization, distributed computing and utility-style service delivery.

VERY IMPORTANT EXAM SUGGESTION: Draw a left-to-right timeline and underline: mainframe, ARPANET, virtual machine, Internet, AWS, EC2 and Azure.

Q2. Explain recent trends in computing: cluster, grid, utility and cloud computing.

Source basis: uploaded lecture PDF, pages 2 | Suggested answer value: 7 marks

Introduction

The lecture presents four stages of computing that progressively increase resource sharing, geographical distribution and service abstraction.

Main Points

1. **Cluster computing:** Homogeneous computing nodes, connected loosely or tightly, work together as one system.
2. **Grid computing:** Heterogeneous nodes distributed over a wide area cooperate to perform very large tasks.
3. **Utility computing:** Computing and storage resources are packaged and supplied like a metered utility.
4. **Cloud computing:** A shared pool of configurable resources is available dynamically and on demand.
5. **Resource organization:** Clusters are usually local and similar; grids are distributed and heterogeneous; cloud hides most infrastructure details.
6. **Access model:** Utility and cloud models emphasize service access, while cluster and grid models emphasize coordinated computation.
7. **Progressive abstraction:** The trend moves from managing machines directly toward consuming flexible services through a network.

Example

A scientific simulation may run on a grid, while a web start-up may obtain servers and storage from a public cloud.

Conclusion

Cloud computing extends earlier distributed and utility concepts by adding ubiquitous, elastic and on-demand access.

VERY IMPORTANT EXAM SUGGESTION: Use a four-stage arrow: Cluster -> Grid -> Utility -> Cloud. A small comparison table can earn an extra presentation mark.

Q3. Define cloud computing and explain the NIST visual model.

Source basis: uploaded lecture PDF, pages 4-5 | Suggested answer value: 7 marks

Introduction

According to the NIST-based lecture definition, cloud computing enables convenient, on-demand network access to a shared pool of configurable computing resources.

Main Points

1. **Shared resources:** Servers, storage, networks and applications are pooled for use by many consumers.
2. **On-demand access:** Resources can be obtained whenever required with minimal provider interaction.
3. **Rapid allocation and release:** Capacity can be provisioned and released quickly with low management effort.
4. **High-level abstraction:** Users consume computation and storage without handling all underlying hardware details.
5. **Essential characteristics:** Broad network access, rapid elasticity, measured service, on-demand self-service and resource pooling.
6. **Service models:** Software as a Service, Platform as a Service and Infrastructure as a Service.
7. **Deployment models:** Public, private, hybrid and community cloud arrangements.

Quick diagram: Essential characteristics -> Service models -> Deployment models

Example

A user opens a cloud application through a browser, while the provider dynamically supplies the required servers and storage.

Conclusion

The NIST visual model describes cloud computing through three dimensions: characteristics, service models and deployment models.

VERY IMPORTANT EXAM SUGGESTION: Start with the definition, then draw three labeled boxes. This structure directly matches the source slide and improves examiner readability.

Q4. Discuss the business advantages and general characteristics of cloud computing.

Source basis: uploaded lecture PDF, pages 6-7 | Suggested answer value: 7 marks

Introduction

Cloud computing provides financial, operational and technical benefits by converting infrastructure into an accessible, shared service.

Main Points

1. **Low upfront investment:** The organization can avoid purchasing a large amount of infrastructure at the beginning.
2. **Real-time availability:** Computing infrastructure can be made available quickly when a business need arises.
3. **Efficient utilization:** Shared resources are allocated according to demand, reducing idle capacity.
4. **Usage-based costing:** Consumers pay according to the amount or duration of resources used.
5. **Reduced time to market:** Applications can be deployed faster because infrastructure provisioning is quicker.
6. **Agility, reliability and scalability:** Resources can be changed dynamically while the system remains dependable.
7. **Ubiquity, multitenancy and lower maintenance:** Services are device/location independent and costs are shared across many users.

Example

A new online store can launch on rented cloud infrastructure without first building a data center.

Conclusion

Cloud computing improves business responsiveness while reducing infrastructure cost, maintenance effort and resource wastage.

VERY IMPORTANT EXAM SUGGESTION: Separate the answer into Business Advantages and General Characteristics. Underline pay-per-use, multitenancy, scalability and agility.

Q5. Explain the five essential characteristics of cloud computing.

Source basis: uploaded lecture PDF, pages 8-12 | Suggested answer value: 7 marks

Introduction

The NIST model identifies five characteristics that distinguish a true cloud service from ordinary remote hosting.

Main Points

1. **Broad network access:** Resources are available over a network through standard mechanisms and heterogeneous clients such as mobiles and laptops.
2. **Rapid elasticity:** Capacity is added or removed rapidly and automatically for scale-out and scale-in.
3. **Measured service:** Usage is recorded, monitored, controlled and optimized transparently for provider and consumer.
4. **On-demand self-service:** Users can obtain server time or storage automatically without continuous human interaction.
5. **Resource pooling:** The provider pools resources to serve multiple users through a multi-tenant model.
6. **Demand-based allocation:** Pooled capacity is assigned according to each consumer's current requirement.
7. **Perception of abundance:** Elasticity makes resources appear practically unlimited to the consumer.

Example

During a traffic surge, a cloud application automatically adds virtual servers, records their usage and releases them later.

Conclusion

Together, these five characteristics provide accessibility, automation, elasticity, sharing and accountable consumption.

VERY IMPORTANT EXAM SUGGESTION: Use the mnemonic B-R-M-O-R: Broad access, Rapid elasticity, Measured service, On-demand service, Resource pooling.

Q6. Explain the components of cloud computing.

Source basis: uploaded lecture PDF, pages 13 | Suggested answer value: 7 marks

Introduction

The cloud computing environment is organized as a layered collection of clients, services, applications, platforms, storage and infrastructure.

Main Points

1. Clients or end-users: Thick clients, thin clients and mobile devices access cloud resources.

2. Services: Products and solutions such as identity, mapping and search are exposed to users or applications.

3. Applications: Web applications and SaaS products deliver complete software functions.

4. Platform: PaaS supports application development, deployment and web hosting.

5. Storage: Databases and Data-Storage-as-a-Service provide persistent data facilities.

6. Infrastructure: Virtualization, IaaS and resources such as EC2 supply the computing foundation.

7. Layer interaction: Upper layers consume facilities of lower layers while hiding their implementation details.

Quick diagram: Clients -> Services -> Applications -> Platform -> Storage -> Infrastructure

Example

A mobile client uses a SaaS application that runs on a PaaS platform and stores data on virtualized infrastructure.

Conclusion

Each component performs a distinct role, but all layers cooperate to deliver cloud services to the end-user.

VERY IMPORTANT EXAM SUGGESTION: Draw the six-layer stack exactly in top-to-bottom order. A correct stack diagram is a strong scoring element.

Q7. Compare SaaS, PaaS and IaaS service models.

Source basis: uploaded lecture PDF, pages 14-17, 28-30, 49 | Suggested answer value: 7 marks

Introduction

Cloud service models differ mainly in the level of abstraction provided and the amount of control retained by the user.

Main Points

1. **SaaS:** The provider delivers a complete application; the user mainly uses and configures the software.
2. **PaaS:** The provider supplies an application development and deployment platform; the user controls deployed applications.
3. **IaaS:** The provider supplies compute, storage, networking and virtualization; the user controls operating systems and software.
4. **Control level:** User control is lowest in SaaS, moderate in PaaS and highest in IaaS.
5. **Management burden:** Provider responsibility increases from IaaS toward SaaS.
6. **Relationship:** The lower service model supports management, computing power and security for the upper model.
7. **Selection basis:** Choose SaaS for ready software, PaaS for development and IaaS for infrastructure flexibility.

Example

Office 365 is SaaS, Windows Azure/App Engine is PaaS, and Amazon EC2 is IaaS in the lecture examples.

Conclusion

SaaS, PaaS and IaaS form a service stack from complete applications down to basic infrastructure.

VERY IMPORTANT EXAM SUGGESTION: Use a three-row comparison with columns: provides, user controls, provider controls and example.

Q8. Explain the architecture of cloud service models and their self-service interfaces.

Source basis: uploaded lecture PDF, pages 28 | Suggested answer value: 7 marks

Introduction

The lecture architecture shows SaaS, PaaS and IaaS as connected layers supported by self-service interfaces and a common administration service.

Main Points

1. **SaaS application layer:** Complete applications are delivered to users at the upper layer.
2. **PaaS self-service interface:** Developers obtain platform functions through a self-service interface.
3. **PaaS facilities:** The source includes security and identity, integration workflow, application grid, UI services and database grid.
4. **IaaS self-service interface:** Infrastructure resources are exposed through a separate self-service interface.
5. **IaaS facilities:** Virtual machines, virtual storage and virtual grids form the infrastructure resource set.
6. **Administration service:** Packaging, configuration, deployment, scaling and lifecycle management support the service stack.
7. **Operational management:** Utilization and user management coordinate access and efficient resource use across the layers.

Quick diagram: Applications -> SaaS -> PaaS -> IaaS; Admin Service supports all layers

Example

A developer uses PaaS tools to deploy an application while IaaS supplies virtual machines and the admin service performs scaling.

Conclusion

The architecture integrates layered services, self-service resource access and centralized administration.

VERY IMPORTANT EXAM SUGGESTION: Draw the central SaaS-PaaS-IaaS stack, then place the PaaS interface on the left, IaaS interface below and Admin Service on the right.

Q9. Explain Software-as-a-Service (SaaS).

Source basis: uploaded lecture PDF, pages 15, 38, 55-57 | Suggested answer value: 7 marks

Introduction

SaaS is a cloud model in which a provider hosts a software application and makes it available to users through the Internet.

Main Points

1. **Application as a service:** The provider's application is delivered as an online service rather than installed and managed locally.
2. **Client access:** Users access the application through a browser, app or other client device.
3. **Provider control:** The provider controls the cloud infrastructure and application operation.
4. **On-demand availability:** The software is available wherever the cloud can be reached through the Internet.
5. **Easy maintenance:** Upgrades and maintenance are handled centrally by the provider.
6. **Scalability and multitenancy:** The service scales up or down and supports multiple users on a shared system.
7. **Cost effectiveness:** Users can pay according to usage and avoid the risk of purchasing unsuitable software.

Example

Microsoft Office 365, Google Apps, Salesforce and Learn.com are examples stated in the slides.

Conclusion

SaaS offers convenient, centrally managed software with minimal technical responsibility for the end-user.

VERY IMPORTANT EXAM SUGGESTION: Underline on-demand, browser access, provider-managed, multitenancy and pay-per-use. Mention at least two examples.

Q10. Explain Platform-as-a-Service (PaaS), its types and benefits.

Source basis: uploaded lecture PDF, pages 16, 38, 52-55 | Suggested answer value: 7 marks

Introduction

PaaS provides a managed environment in which consumers can develop, deploy and run their own or acquired applications.

Main Points

1. **Development and deployment support:** The platform supplies tools and runtime facilities for application creation and deployment.
2. **User control:** The user controls deployed applications and available configuration settings, but not the underlying infrastructure.
3. **Application delivery-only environment:** It provides on-demand scaling and application security.
4. **Stand-alone development environment:** It offers an independent platform for a specific development function.
5. **Open PaaS and add-on facilities:** Open-source platforms run applications, while add-ons customize existing SaaS platforms.
6. **Lower overhead and ownership cost:** Users avoid platform administration and do not need to purchase servers or memory.
7. **Scalable and current software:** Resources scale automatically and the provider maintains upgrades and patches.

Example

Windows Azure and Google App Engine are PaaS examples in the lecture slides.

Conclusion

PaaS accelerates software development by supplying a scalable, maintained platform while hiding infrastructure complexity.

VERY IMPORTANT EXAM SUGGESTION: For full marks, include definition, four PaaS types, two benefits and one example.

Q11. Explain Infrastructure-as-a-Service (IaaS) and its working methodology.

Source basis: uploaded lecture PDF, pages 17, 30-31, 50-52 | Suggested answer value: 7 marks

Introduction

IaaS is the on-demand delivery of basic computing infrastructure through the Internet.

Main Points

1. **Resources supplied:** It provides compute servers, storage, networking, operating-system support and virtualization features.
2. **Virtual-machine access:** Administrative access to VMs enables users to control computing resources.
3. **User responsibility:** The user can deploy, execute and control operating systems and other software.
4. **Selected network control:** In some cases, the user controls components such as host firewalls, IP addresses and VLANs.
5. **Working method:** The user rents servers, data-center space or network equipment; the provider performs resource management.
6. **Virtualized server:** A hypervisor divides physical hardware into multiple VMs, each with its own operating system and application.
7. **On-demand outsourced service:** Capacity is rented by resource type, usage time or service package.

Example

Amazon EC2, GoGrid, iland and Rackspace Cloud Servers are examples listed in the source.

Conclusion

IaaS gives users high flexibility and administrative control without requiring ownership of physical infrastructure.

VERY IMPORTANT EXAM SUGGESTION: Draw Physical Server versus Virtualized Server with Hardware -> Hypervisor -> VM1/VM2. This diagram is highly important.

Q12. Explain cloud deployment models.

Source basis: uploaded lecture PDF, pages 18 | Suggested answer value: 7 marks

Introduction

Deployment models specify who owns, manages and accesses a cloud environment and where its infrastructure is located.

Main Points

1. Public cloud: A provider offers cloud services for use by the general public or many organizations.

2. Private cloud: The cloud is dedicated to a single organization and may be managed internally or by a third party.

3. Hybrid cloud: Two or more distinct cloud environments are connected to support portability and combined operation.

4. Community cloud: Several organizations with common concerns share one cloud environment.

5. Distributed cloud: Computing devices are scattered across locations but connected through a single network.

6. Multi-cloud: Multiple cloud services are used through a heterogeneous architecture for flexibility and fault tolerance.

7. Inter-cloud: A global cloud-of-clouds concept supports interoperability among cloud service providers.

Example

A company may keep sensitive data in a private cloud while using a public cloud for temporary extra capacity.

Conclusion

The correct deployment model depends on access, control, security, cost and interoperability requirements.

VERY IMPORTANT EXAM SUGGESTION: Begin with public, private and hybrid, then write the four other types. Do not confuse multi-cloud with hybrid cloud.

Q13. Explain public cloud and private cloud, and compare them.

Source basis: uploaded lecture PDF, pages 19-21 | Suggested answer value: 7 marks

Introduction

Public and private clouds differ mainly in ownership, sharing, customer scope, connectivity and security control.

Main Points

1. **Public cloud meaning:** It is offered for use by any person or industry and is normally owned by a cloud service organization.
2. **Public cloud benefits:** It is easy to set up, scalable and pay-per-use because the provider covers hardware, application and bandwidth costs.
3. **Private cloud meaning:** It serves only one organization and may operate on-premises or off-premises.
4. **Private cloud benefits:** It provides greater control over systems and data and reduces security concerns.
5. **Private cloud limitation:** The organization or its contractor must perform regular maintenance.
6. **Resource sharing:** Public resources are publicly shared among multiple customers; private resources are shared within a limited group.
7. **Connectivity and security:** Public cloud mainly uses the Internet with lower control; private cloud may use private networks and offers higher security control.

Example

AWS, Google Compute Engine and Microsoft Azure are public-cloud examples; an internal enterprise cloud is private.

Conclusion

Public cloud favors cost and scalability, whereas private cloud favors dedicated control and security.

VERY IMPORTANT EXAM SUGGESTION: Use a two-column table with rows: resources, customers, connectivity, security, advantages and limitation.

Q14. Explain hybrid, community, distributed, multi-cloud and inter-cloud.

Source basis: uploaded lecture PDF, pages 22-24 | Suggested answer value: 7 marks

Introduction

Beyond basic public and private clouds, several models combine or distribute resources to meet specialized needs.

Main Points

1. **Hybrid cloud:** Two or more unique clouds are joined through standardized tools.
2. **Hybrid portability:** It supports data and application portability, including load balancing between clouds.
3. **Community cloud:** Organizations with common security, compliance or jurisdiction concerns share a cloud.
4. **Community management:** It may be managed internally by participating organizations or by a third party.
5. **Distributed cloud:** Scattered computing devices in different locations are connected to a single network.
6. **Multi-cloud:** Multiple cloud services are offered through a heterogeneous architecture, increasing flexibility and fault tolerance.
7. **Inter-cloud:** A unified cloud-of-clouds supports interoperability between different cloud providers.

Example

A regulated group of hospitals may share a community cloud, while an enterprise may use several providers in a multi-cloud setup.

Conclusion

These models extend cloud deployment through sharing, distribution, integration and provider interoperability.

VERY IMPORTANT EXAM SUGGESTION: Write one distinguishing phrase for each model. The key words are portability, common concerns, scattered devices, multiple providers and cloud-of-clouds.

Q15. Compare different cloud deployment models using location and access.

Source basis: uploaded lecture PDF, pages 25 | Suggested answer value: 7 marks

Introduction

The source classifies deployment models using two dimensions: infrastructure location and whether access is dedicated or shared.

Main Points

1. **On-premise plus dedicated access:** This combination produces a private cloud controlled for one organization.
2. **Off-premise plus dedicated access:** This produces a hosted private cloud managed outside the organization.
3. **On-premise plus shared access:** This corresponds to a community cloud shared by related organizations.
4. **Off-premise plus shared access:** This corresponds to a public cloud offered by a provider.
5. **Location dimension:** On-premise offers direct organizational placement; off-premise shifts infrastructure to a service provider.
6. **Access dimension:** Dedicated access emphasizes exclusivity; shared access emphasizes resource pooling among consumers.
7. **Decision factors:** Security, cost, maintenance, customer scope and control determine the best combination.

Example

A bank may select an on-premise private cloud, while a small business may select an off-premise public cloud.

Conclusion

The location-access matrix provides a simple method for distinguishing private, hosted private, community and public clouds.

VERY IMPORTANT EXAM SUGGESTION: Reproduce the 2 x 2 matrix exactly; it is compact and easy for an examiner to award marks.

Q16. Compare the responsibility of user and service provider in traditional IT, IaaS, PaaS and SaaS.

Source basis: uploaded lecture PDF, pages 29 | Suggested answer value: 7 marks

Introduction

As the service model moves from traditional IT to SaaS, more system layers are managed by the cloud service provider.

Main Points

1. Traditional IT: The user manages applications, data, runtime, middleware, OS, virtualization, servers, storage and networking.

2. IaaS provider role: The provider manages networking, storage, servers and virtualization.

3. IaaS user role: The user manages the OS, middleware, runtime, data and applications.

4. PaaS provider role: The provider additionally manages the OS, middleware and runtime.

5. PaaS user role: The user mainly manages applications and data.

6. SaaS provider role: The provider manages the complete stack from networking to applications.

7. Control-management trade-off: More provider management gives greater convenience but less direct user control.

Example

A developer needing OS control selects IaaS; a developer needing only a runtime selects PaaS; an office user selects SaaS.

Conclusion

Responsibility progressively shifts from the user to the provider in the order Traditional -> IaaS -> PaaS -> SaaS.

VERY IMPORTANT EXAM SUGGESTION: Draw a nine-layer stack and shade user-managed versus provider-managed sections.

Q17. Discuss IaaS procedures and benefits.

Source basis: uploaded lecture PDF, pages 50-52 | Suggested answer value: 7 marks

Introduction

IaaS enables users to obtain virtualized infrastructure, administer it and pay according to the rented resources.

Main Points

1. Request and allocation: A user requests infrastructure and receives a virtual machine or related resources.

2. Administrative access: VM administration permits saving data, starting a web server and installing applications.

3. Flexible hardware renting: Virtual desktops and server resources can be rented without purchasing hardware.

4. Wide resource range: Rentable items include VMs, storage, bandwidth, firewalls, IP addresses and monitoring services.

5. Usage-based payment: Charges depend on resource type, usage time and selected service package.

6. Portability: Infrastructure accessed through Internet connections can support movement of workloads.

7. Legacy interoperability: IaaS can maintain interoperability with existing applications and workloads across IaaS clouds.

Example

A software owner creates a VM, installs a web server, stores data and grants access to multiple users.

Conclusion

IaaS combines administrative flexibility, resource rental, portability and compatibility with existing workloads.

VERY IMPORTANT EXAM SUGGESTION: Mention at least three VM commands and four rentable resources to make the answer specific.

Q18. Explain SaaS characteristics and advantages.

Source basis: uploaded lecture PDF, pages 55-57 | Suggested answer value: 7 marks

Introduction

SaaS delivers complete software through the cloud and centralizes operation, maintenance and data management.

Main Points

1. **Internet delivery:** Software is deployed on a cloud server and accessed through the Internet.
2. **On-demand availability:** The service can be used from any location where the cloud is reachable.
3. **Easy maintenance:** Users do not perform upgrades or routine software maintenance.
4. **Flexible scaling:** Centralized management enables the service to scale up or down.
5. **Shared data model:** Multiple users can share a single data model and database.
6. **Cost effectiveness:** Payment is usage-based and users avoid purchasing unsuitable software.
7. **Multitenant consistency:** Multiple programmers or users operate on the same software version, reducing version mismatch.

Example

A cloud email or office suite is used by many users while the provider maintains one centrally managed service.

Conclusion

SaaS reduces installation and maintenance effort while providing scalable, shared and economical application access.

VERY IMPORTANT EXAM SUGGESTION: Do not repeat only the definition. Explain on-demand, maintenance, scaling, shared data, cost and multitenancy.

Q19. Explain virtualization and its benefits in cloud computing.

Source basis: uploaded lecture PDF, pages 31, 39 | Suggested answer value: 7 marks

Introduction

Virtualization makes one physical computer appear as multiple independent computers called virtual machines.

Main Points

1. **Virtual-machine concept:** Each VM behaves like a separate computer with its own operating system and applications.
2. **Hypervisor:** A hypervisor sits above hardware and creates, isolates and manages multiple VMs.
3. **Resource pooling:** CPU, memory, storage and network capacity are treated as a pool for virtual allocation.
4. **Sharing and cost reduction:** Several workloads share the same physical resource, improving utilization.
5. **Encapsulation:** A VM contains a complete computing environment in a manageable package.
6. **Hardware independence:** The VM operates independently of the exact underlying physical hardware.
7. **Portability:** VM migration enables a virtual environment to move between physical hosts.

Example

A single server can host VM1 for a web application and VM2 for a database, each with a separate operating system.

Conclusion

Virtualization is a core cloud-enabling technology because it provides isolation, flexibility, portability and efficient sharing.

VERY IMPORTANT EXAM SUGGESTION: Draw Hardware -> Hypervisor -> VM1 and VM2. Underline sharing, encapsulation, independence and portability.

Q20. Explain Wireless Sensor Networks (WSNs), sensor-node components and applications.

Source basis: uploaded lecture PDF, pages 34-35 | Suggested answer value: 7 marks

Introduction

A WSN contains sensor nodes that observe physical phenomena and communicate sensed data to a centralized sink node.

Main Points

1. **Sensing:** Sensor nodes measure environmental or physical conditions.
2. **Wireless transmission:** The sensed data is transmitted through wireless communication.
3. **Sink node:** A centralized unit called the sink receives data from sensor nodes.
4. **Single-hop or multi-hop communication:** A node may communicate directly with the sink or forward data through other nodes.
5. **Data processing:** The sink performs further processing or aggregation of collected data.
6. **Sensor-node components:** A node mainly contains a sensing unit, processing unit and communication unit.
7. **Applications:** Target tracking, wildlife monitoring, healthcare, industry, smart homes, smart cities and agriculture.

Example

Agricultural sensor nodes measure field conditions and send readings through multiple hops to a sink for analysis.

Conclusion

A WSN combines distributed sensing, local processing and wireless communication to support monitoring applications.

VERY IMPORTANT EXAM SUGGESTION: Draw five sensor nodes connected to one sink and label the three units inside a sensor node.

Q21. Discuss the limitations of traditional WSNs.

Source basis: uploaded lecture PDF, pages 40 | Suggested answer value: 7 marks

Introduction

Traditional WSN deployment creates procurement, deployment, maintenance and requirement-change difficulties for users.

Main Points

1. **Price:** Purchasing sensor nodes and related infrastructure can require significant investment.
2. **Vendor selection:** The user must identify a suitable and dependable sensor vendor.
3. **Sensor-type selection:** The correct physical sensors must be integrated for the intended application.
4. **Deployment method:** Sensors must be installed in the correct way to obtain useful readings.
5. **Deployment location:** Incorrect placement can reduce coverage, accuracy or connectivity.
6. **Post-deployment maintenance:** Faults, environmental effects and network changes require continued support.
7. **Battery and changing requirements:** Limited battery life and new applications may force changes in sensor type, area and topology.

Example

Changing from an agriculture application to a smart-home application may require different sensors, deployment area and topology.

Conclusion

WSN ownership places technical and financial responsibility on the user and is difficult to adapt when requirements change.

VERY IMPORTANT EXAM SUGGESTION: Organize the answer under Procurement, Deployment, Maintenance and Change of Requirement.

Q22. Define sensor-cloud and explain its main features.

Source basis: uploaded lecture PDF, pages 33, 36-41 | Suggested answer value: 7 marks

Introduction

Sensor-cloud is more than merely connecting sensors to a cloud or dumping sensor data into cloud storage; it virtualizes sensor resources as services.

Main Points

1. **Sensor virtualization:** A physical sensor node or network is represented through virtual sensor resources.
2. **One-to-many appearance:** One physical sensor or WSN can appear as multiple logical resources for different users.
3. **Sensor-as-a-Service:** End-users consume sensing capabilities as Se-aaS through applications.
4. **Pay-per-use:** Consumers pay according to their use of sensor resources and services.
5. **Intermediary stratum:** A sensor-cloud layer is placed between physical sensor nodes and end-users.
6. **Cloud properties:** Elasticity, self-service and shared resource use are applied to sensing infrastructure.
7. **Multiple applications:** The same physical sensing infrastructure can serve several applications or organizations.

Example

A temperature-sensor network can provide separate virtual data services to agriculture and environmental-monitoring applications.

Conclusion

Sensor-cloud converts dedicated sensor networks into virtualized, shareable and service-oriented sensing infrastructure.

VERY IMPORTANT EXAM SUGGESTION: Begin by writing what sensor-cloud is NOT, then explain virtualization, Se-aaS, pay-per-use and multiple users.

Q23. Differentiate WSN and sensor-cloud.

Source basis: uploaded lecture PDF, pages 42-43 | Suggested answer value: 7 marks

Introduction

A traditional WSN is normally dedicated to one user, while sensor-cloud virtualizes physical sensors for multiple applications and users.

Main Points

1. **Usage model:** WSN serves a dedicated user; sensor-cloud serves multiple end-users or applications.
2. **Virtualization:** Traditional WSN does not provide sensor virtualization; sensor-cloud creates virtual sensor instances.
3. **Ownership:** In WSN, the WSN user owns the network; in sensor-cloud, a sensor-owner owns physical devices.
4. **Deployment:** WSN deployment is performed by the WSN user; sensor-cloud deployment is performed by the sensor-owner.
5. **Redeployment and maintenance:** The WSN user handles these in WSN, while the Sensor-Cloud Service Provider handles them in sensor-cloud.
6. **Overhead:** WSN overhead lies with the user; sensor-cloud overhead is shifted to the SCSP.
7. **Data delivery:** WSN provides aggregated data to one user; sensor-cloud provisions sensed services through cloud infrastructure.

Example

Instead of each organization installing a separate tracking WSN, several organizations can request virtual tracking services.

Conclusion

Sensor-cloud separates ownership, management and usage, reducing user overhead and increasing sharing.

VERY IMPORTANT EXAM SUGGESTION: Use a two-column table and include the six actor-role rows: ownership, deployment, redeployment, maintenance, overhead and usage.

Q24. Explain the actors and their roles in sensor-cloud.

Source basis: uploaded lecture PDF, pages 43-45 | Suggested answer value: 7 marks

Introduction

Sensor-cloud operation depends on three main business and technical actors: end-users, sensor-owners and the Sensor-Cloud Service Provider.

Main Points

1. **End-user role:** The end-user consumes Sensor-as-a-Service through applications according to requirements.
2. **End-user abstraction:** The user need not know which physical sensor is allocated to the application.
3. **End-user request:** Users register, select templates and request applications or sensing services.
4. **Sensor-owner role:** The sensor-owner purchases physical sensor devices and considers their business use.
5. **Physical deployment:** The sensor-owner deploys homogeneous or heterogeneous nodes in different geographical locations.
6. **SCSP role:** The SCSP performs managerial functions, resource coordination, redeployment and maintenance.
7. **Charging:** The SCSP charges end-users according to their use of Se-aaS.

Example

A city department requests a pollution-monitoring template, while a sensor-owner supplies nodes and the SCSP allocates them.

Conclusion

The actor separation enables users to consume sensing services without owning or directly maintaining the sensor network.

VERY IMPORTANT EXAM SUGGESTION: Write each actor as a separate subheading and state one business role plus one technical role.

Q25. Explain the sensor-cloud architecture and views.

Source basis: uploaded lecture PDF, pages 45-46 | Suggested answer value: 7 marks

Introduction

Sensor-cloud architecture connects user organizations, a browser/web portal, cloud management functions and a heterogeneous pool of physical sensors.

Main Points

1. **User organization view:** Applications receive data feeds while users interact through a browser interface.
2. **Registration and login:** The user authenticates before requesting a sensing application.
3. **Template display and specification:** The portal displays templates and receives application-specific requirements.
4. **Web portal:** The portal interprets specifications and exchanges sensed data with the cloud infrastructure.
5. **Cloud management:** Functions include dynamic scaling, on-demand physical-sensor scheduling, storage and specialized processing.
6. **QoS and energy management:** The architecture manages energy, quality of service and application-specific real-time aggregation.
7. **Physical-sensor pool:** Heterogeneous sensors provide on-demand data through controlled interaction with the cloud layer.

Quick diagram: User/Application <-> Browser Interface <-> Web Portal/Cloud <-> Physical Sensor Pool

Example

A user selects a target-tracking template; the cloud schedules suitable sensors and returns an aggregated data feed.

Conclusion

The architecture hides physical sensor complexity while providing scalable, application-specific sensing services.

VERY IMPORTANT EXAM SUGGESTION: Draw the four-block flow and label template specification in one direction and sensed data in the reverse direction.

Q26. Explain the workflow of a sensor-cloud system.

Source basis: uploaded lecture PDF, pages 47 | Suggested answer value: 7 marks

Introduction

The sensor-cloud workflow creates a virtual sensor from a user request, allocates physical resources, provides data and releases resources when finished.

Main Points

1. **Operations request:** The user organization sends an operation request to the SensorML interpreter.
2. **Template interpretation:** The interpreter decodes the XML/SensorML template containing sensor, group, client and metadata definitions.
3. **Virtual instance creation:** A Virtual Sensor Manager creates the required virtual sensor instance.
4. **Operation control:** The Virtual Sensor Controller manages the requested virtual-sensor operations.
5. **Resource scheduling:** The Resource Manager finds compatible sensors and performs allocation or deallocation from the WSN pool.
6. **Data retrieval and aggregation:** Sensor data is retrieved, aggregated and provisioned to the user organization.
7. **Deletion and release:** When the service ends, the virtual sensor is deleted and the instance and physical resources are released.

Example

A data request for temperature monitoring creates a virtual sensor, allocates compatible nodes, aggregates readings and later releases them.

Conclusion

The workflow provides complete lifecycle management from request and allocation to data delivery and resource release.

VERY IMPORTANT EXAM SUGGESTION: Write the sequence as arrows: User -> Interpreter -> Manager -> Controller -> Resource Manager -> Sensor Pool.

Q27. Explain the target-tracking case study in sensor-cloud.

Source basis: uploaded lecture PDF, pages 48 | Suggested answer value: 7 marks

Introduction

The case study compares costly dedicated WSN ownership with service-based target tracking in a sensor-cloud environment.

Main Points

1. Traditional requirement: An organization wishing to detect intrusion in a zone normally needs a target-tracking WSN.

2. Data-sharing problem: A WSN owner may refuse to share sensed information with an external organization.

3. Duplicate deployment: Each interested organization then has to deploy its own WSN.

4. Long-term investment: Network installation creates substantial setup cost.

5. Maintenance overhead: The organization must also bear continued maintenance responsibility.

6. Sensor-cloud alternative: The same tracking application can be consumed as a service without owning the WSN.

7. Economic effect: Shared sensor infrastructure reduces duplication and converts ownership cost into service usage.

Example

A security agency requests intrusion-detection data from a sensor-cloud instead of installing a new sensor network in the zone.

Conclusion

Sensor-cloud makes target tracking more economical and accessible by separating application use from physical WSN ownership.

VERY IMPORTANT EXAM SUGGESTION: Present the answer as Problem -> Traditional consequence -> Sensor-cloud solution -> Benefit.

Q28. Summarize the historical cloud market analysis presented in the lecture.

Source basis: uploaded lecture PDF, pages 60-63 | Suggested answer value: 7 marks

Introduction

The lecture provides a historical 2017 snapshot of cloud infrastructure growth and identifies the major provider companies of that period.

Main Points

1. Historical perspective: The slides connect the evolution from mainframes to cloud-based infrastructure services.

2. SaaS business interest: A referenced 2016 article lists business reasons for adopting SaaS.

3. 2017 market snapshot: The lecture uses 2017 market analyses to illustrate rapid growth of cloud infrastructure services.

4. Amazon leadership: The source labels Amazon as the leading cloud provider in the market snapshot.

5. Provider concentration: The cited analysis states that leading cloud providers were increasing their market separation.

6. Major companies: The lecture names Amazon Web Services, Microsoft, IBM and Google as top cloud-provider companies.

7. Study caution: These rankings are historical slide content and should be written with the year 2017, not as a current claim.

Example

An exam answer can describe the 2017 slide as showing AWS ahead, followed by other large providers.

Conclusion

The historical analysis demonstrates the commercialization and concentration of cloud services among major providers.

VERY IMPORTANT EXAM SUGGESTION: Always write “as presented in the 2017 lecture snapshot.” This prevents an outdated chart from being stated as present-day fact.

Q29. Explain AWS and its major cloud services described in the lecture.

Source basis: uploaded lecture PDF, pages 63-67 | Suggested answer value: 7 marks

Introduction

AWS is presented as a major global provider offering cloud products across computing, storage, databases, networking, IoT, management and security.

Main Points

1. Service breadth: AWS supplies computing, storage, databases, analytics, networking, mobile, IoT and enterprise tools.

2. Amazon EC2: EC2 provides secure and resizable cloud computing capacity.

3. Developer support: EC2 tools support failure-resilient application development and help remove common failure scenarios.

4. Amazon S3: S3 provides object storage through web-service interfaces.

5. S3 access and security: Access can be controlled and monitored, including secure connections through VPC endpoints.

6. S3 interfaces: The source lists REST, SOAP and BitTorrent interfaces for storage access.

7. Amazon Cloud Drive: The lecture describes web storage, multi-device access, music playback and storage options.

Example

A web application can run on EC2 and store files in S3 with controlled access.

Conclusion

AWS combines elastic computing and managed storage with security, interfaces and development support.

VERY IMPORTANT EXAM SUGGESTION: Focus on EC2 and S3. Write full forms: Elastic Compute Cloud and Simple Storage Service.

Q30. Explain Microsoft cloud services described in the lecture.

Source basis: uploaded lecture PDF, pages 68-69 | Suggested answer value: 7 marks

Introduction

The lecture describes Microsoft's shift toward cloud technology through Office 365 and Microsoft Azure.

Main Points

1. **Cloud focus:** The source notes that Microsoft reduced emphasis on hardware development and focused more on cloud technology.
2. **Office 365:** A cloud-based subscription service combines applications such as Excel and Outlook with online services.
3. **Collaboration services:** OneDrive and Microsoft Teams support creation and sharing across devices.
4. **Microsoft Azure:** Azure provides compute, networking, storage, web, mobile, database and data-analysis services.
5. **Application lifecycle:** Users can access, build and deploy cloud solutions where needed.
6. **Hybrid support:** Azure supports hybrid cloud technology.
7. **Security and intelligent applications:** The platform supports data-driven applications and emphasizes security, privacy, transparency and compliance.

Example

An organization can use Office 365 for collaboration and Azure for deploying a data-driven web application.

Conclusion

Microsoft combines end-user SaaS with a broad cloud platform for application and hybrid-cloud deployment.

VERY IMPORTANT EXAM SUGGESTION: Differentiate Office 365 as SaaS from Azure as a broad cloud platform.

Q31. Explain IBM cloud services described in the lecture.

Source basis: uploaded lecture PDF, pages 70-71 | Suggested answer value: 7 marks

Introduction

The lecture presents IBM as a provider of public, private and hybrid cloud services across IaaS, PaaS and SaaS.

Main Points

1. **Multi-model support:** IBM supports public, private and hybrid cloud environments.
2. **Service portfolio:** The source states that IBM offers more than 130 services across IaaS, PaaS and SaaS.
3. **Cloud Object Storage:** It supports data storage, management and controlled access.
4. **Integration:** Applications can connect directly to object storage and integrate with other IBM cloud services.
5. **Interfaces:** Self-service portals and REST APIs provide access and management.
6. **IBM Cloud Private:** A pre-packaged enterprise platform supports building or modernizing applications with security, speed and control.
7. **Streaming Video:** IBM provides workflow-management and B2B video-streaming solutions.

Example

An enterprise may store objects through REST APIs and deploy a controlled internal application on IBM Cloud Private.

Conclusion

IBM's cloud portfolio emphasizes enterprise integration, controlled storage, private-cloud operation and managed services.

VERY IMPORTANT EXAM SUGGESTION: Remember the three named services: Cloud Object Storage, Cloud Private and Cloud Streaming Video.

Q32. Explain Google Cloud services described in the lecture.

Source basis: uploaded lecture PDF, pages 72-74 | Suggested answer value: 7 marks

Introduction

Google Cloud Platform provides compute, application hosting, storage, analytics, API and database services.

Main Points

1. Google Compute Engine: It provides IaaS-based virtual machines in Google data centers with load-balancing support.

2. Google App Engine: A PaaS for web applications with automatic scaling according to resource demand and server load.

3. Google Cloud Storage: An online file-storage service.

4. BigQuery: A data-analysis tool that uses SQL-like queries to process large datasets.

5. Cloud Endpoints: A tool for creating services inside App Engine and connecting Android, iOS and JavaScript clients.

6. Cloud DNS: A domain-name service supported by Google Cloud.

7. Datastore and Cloud SQL: Datastore provides NoSQL storage, while Cloud SQL is a managed MySQL database.

Example

A web application can run on App Engine, store objects in Cloud Storage and analyze data using BigQuery.

Conclusion

Google Cloud combines IaaS, PaaS, storage, data analytics and managed database services.

VERY IMPORTANT EXAM SUGGESTION: Classify each product: Compute Engine-IaaS, App Engine-PaaS, BigQuery-analytics, Datastore-NoSQL and Cloud SQL-MySQL.

Q33. Explain Apple iCloud and its major features described in the lecture.

Source basis: uploaded lecture PDF, pages 75-78 | Suggested answer value: 7 marks

Introduction

iCloud is an Apple cloud-storage and cloud-computing service designed to synchronize, protect and access user information across devices.

Main Points

1. **Platform support:** The lecture lists support for Apple operating systems and compatible Microsoft Windows versions.
2. **Service transition:** iCloud replaced Apple's earlier MobileMe online-service collection.
3. **Personal information:** Email, contacts and calendars can be synchronized.
4. **Backup and location:** Backup and Restore, Find My Friends and Find My iPhone support device data and location functions.
5. **Productivity and remote access:** iWork for iCloud and Back to My Mac provide web productivity and remote login functions.
6. **Photos and files:** Photo Stream, iCloud Photo Library and iCloud Drive store photos, videos, documents and app data.
7. **Security and media:** iCloud Keychain manages passwords/cards, while iTunes Match supports music-library matching.

Example

A user can back up an iPhone, locate it if lost and access documents through iCloud Drive.

Conclusion

iCloud integrates storage, synchronization, backup, security, productivity and device-management services for Apple users.

VERY IMPORTANT EXAM SUGGESTION: Group features under synchronization, backup/location, files/photos, security and media.

Q34. Define fog computing and explain its characteristics.

Source basis: uploaded lecture PDF, pages 81-85 | Suggested answer value: 7 marks

Introduction

Fog computing moves storage, communication, control and computing functions from a distant cloud toward end-user or near-user edge devices.

Main Points

1. **Near-user operation:** It uses collaborative end-user clients or edge devices located close to data sources.
2. **Local storage:** A substantial amount of data can be stored near users instead of only in a cloud data center.
3. **Local communication:** Traffic can be handled locally rather than always routed through the Internet backbone.
4. **Local management:** Control, configuration, measurement and management can occur near the edge.
5. **Decentralized infrastructure:** Fog Computing Nodes are distributed between end devices and the cloud.
6. **Heterogeneous nodes:** Routers, switches, access points, IoT gateways and set-top boxes can act as fog nodes.
7. **Fog abstraction layer:** A uniform layer supports allocation, monitoring, security, device management, storage and compute services.

Example

An IoT gateway can process sensor data locally and send only summarized information to the central cloud.

Conclusion

Fog computing reduces dependence on distant clouds by providing decentralized, heterogeneous and near-user services.

VERY IMPORTANT EXAM SUGGESTION: Underline FCN and list at least four FCN examples. Mention both IP and non-IP access support.

Q35. Explain Mobile Edge Computing (MEC), its characteristics and controller.

Source basis: uploaded lecture PDF, pages 86-90 | Suggested answer value: 7 marks

Introduction

MEC places cloud computing and IT services at the edge of a mobile cellular network.

Main Points

1. **Edge location:** MEC servers are usually co-located with a radio network controller or macro base station.
2. **Faster service:** Cloud functions are closer to mobile user equipment, reducing response time.
3. **Congestion reduction:** Local processing reduces traffic that must travel through the wider network.
4. **Reliability:** Near-edge operation supports more reliable applications.
5. **Open RAN access:** Authorized third parties such as application developers or content providers can use radio-network capabilities.
6. **Real-time context:** MEC provides information about network load, capacity, connected devices and location.
7. **MEO controller:** The Mobile Edge Orchestrator manages hosts, resources, topology, information flow and edge applications.

Example

A MEC server at an LTE eNB can process a low-latency mobile application and provide local network-load information.

Conclusion

MEC combines cellular-network context with virtualized computing and storage near the base station.

VERY IMPORTANT EXAM SUGGESTION: Write full forms: MEC, RAN, RNC, eNB/gNB and MEO. The controller functions are frequently examinable.

Q36. Define cloudlet and explain its three-layer architecture.

Source basis: uploaded lecture PDF, pages 91-94 | Suggested answer value: 7 marks

Introduction

A cloudlet is a mobility-enhanced, small-scale cloud data center located near mobile devices at the edge of the Internet.

Main Points

1. **Purpose:** It supports resource-intensive and interactive mobile applications.
2. **Low latency:** Powerful resources are provided near the user through direct, short-distance access.
3. **UE-cloud relationship:** It operates between user equipment and the distant central cloud.
4. **VM provisioning:** Cloudlets use virtual machines to provision resources to user equipment in real time, commonly through Wi-Fi.
5. **Cloudlet layer:** A group of co-located nodes is managed by a Cloudlet Agent.
6. **Node layer:** Multiple execution environments run above the operating system and are managed by a Node Agent.
7. **Component layer:** A set of services interfaces with the higher execution environment.

Example

A nearby cloudlet can execute image processing for a mobile device and return the result with low delay.

Conclusion

Cloudlets bring VM-based computing close to mobile users through a compact three-layer edge architecture.

VERY IMPORTANT EXAM SUGGESTION: Draw UE <-> Cloudlet <-> Cloud and list Cloudlet, Node and Component layers in order.

Q37. Compare fog computing, MEC and cloudlet computing.

Source basis: uploaded lecture PDF, pages 94-95 | Suggested answer value: 7 marks

Introduction

Fog, MEC and cloudlet computing all move resources closer to users, but differ in node type, location, architecture, context awareness and access.

Main Points

1. **Node device:** Fog uses routers, switches, access points and gateways; MEC uses servers at base stations; cloudlet is a data center in a box.
2. **Location:** Fog nodes may exist anywhere between devices and cloud; MEC is tied to radio-network/base-station locations; cloudlets are locally installed near users.
3. **Architecture:** Fog uses a fog abstraction layer, MEC uses a mobile orchestrator and cloudlet uses a cloudlet agent.
4. **Context awareness:** The source rates MEC high, fog medium and cloudlet low.
5. **Proximity:** Fog may be one or multiple hops away; MEC and cloudlet are generally one hop away.
6. **Access:** Fog supports Bluetooth, Wi-Fi and mobile networks; MEC uses mobile networks; cloudlet mainly uses Wi-Fi.
7. **Special strengths:** Fog supports non-IP access, MEC provides strong cellular context, and cloudlet ensures close logical proximity.

Example

An IoT gateway is fog, a server beside a 5G base station is MEC, and a nearby Wi-Fi mini-data-center is a cloudlet.

Conclusion

The best approach depends on network type, context awareness, latency, access method and required proximity.

VERY IMPORTANT EXAM SUGGESTION: Use a comparison table with the exact rows: device, location, architecture, context, proximity, access and inter-node communication.

Q38. Explain the trends and issues that motivate mobile and IoT cloud computing.

Source basis: uploaded lecture PDF, pages 97-99 | Suggested answer value: 7 marks

Introduction

New mobile and IoT applications require high data rates, quality of service and processing capability beyond what many individual devices can supply.

Main Points

1. Growing demand: User requirements for high data rates and QoS are increasing rapidly.

2. Advanced devices: Smartphones, IoT devices, sensors and actuators enable more sophisticated services.

3. Improved CPUs: Modern devices contain stronger processors, but local computing remains limited.

4. Time constraint: Device CPUs may not process all data and return results within the required time.

5. Situational awareness: Optimal decisions require comprehensive information that may not exist at one node.

6. Large data volume: Collected datasets can be too large for a single node to compute efficiently.

7. Battery limitation: Energy constraints restrict sustained high-speed and large-scale local computation.

Example

A mobile device performing complex real-time analysis may offload processing because local energy and computation are insufficient.

Conclusion

These limitations motivate cloud and edge assistance for mobile and IoT computing.

VERY IMPORTANT EXAM SUGGESTION: Connect every issue to a need: limited CPU -> offloading, big data -> cloud storage, battery -> remote computation, latency -> edge.

Q39. Explain Mobile Cloud Computing (MCC) and its advantages.

Source basis: uploaded lecture PDF, pages 99-100 | Suggested answer value: 7 marks

Introduction

MCC allows mobile and IoT users to access computing and storage resources of a distant centralized cloud through the operator core network and Internet.

Main Points

1. **Participants:** User equipment and IoT systems act as clients of the mobile cloud.
2. **Communication path:** Requests travel through a mobile operator core network and the Internet to a centralized cloud.
3. **Remote computation:** Energy-consuming or complex tasks can be offloaded from the device.
4. **Battery extension:** Offloading reduces local processing and can extend battery lifetime.
5. **Advanced application support:** The cloud enables sophisticated SaaS, PaaS and IaaS functions for mobile users.
6. **Massive storage:** Mobile and IoT systems gain access to large cloud-storage capacity.
7. **Central resource strength:** The centralized cloud offers powerful processing and broad resource availability.

Example

A smartphone sends a computation-heavy task through LTE to a cloud server and receives the processed result.

Conclusion

MCC expands mobile-device capability by borrowing centralized cloud computation, services and storage.

VERY IMPORTANT EXAM SUGGESTION: Draw UE/IoT -> LTE/Core Network -> Internet -> Central Cloud and write the three advantages as numbered points.

Q40. Compare conventional MCC and edge computing.

Source basis: uploaded lecture PDF, pages 101-102 | Suggested answer value: 7 marks

Introduction

Conventional MCC uses distant centralized clouds, while edge computing moves computing and storage closer to the user equipment.

Main Points

1. **Deployment:** MCC is centralized; edge computing is distributed near base stations or access points.
2. **Distance:** MCC resources are far from the user; edge resources are close.
3. **Latency:** MCC generally has long delay; edge computing provides short delay.
4. **Jitter and QoS:** MCC can have high jitter and delay-bound violations; edge computing supports more stable high-QoS service.
5. **Network and battery use:** MCC consumes more network resources and device energy; edge reduces wide-area traffic and battery use.
6. **Resource capacity:** MCC has abundant computing and storage; edge capacity is more limited.
7. **Complexity:** Edge systems require prediction, pre-fetching and distributed control functions.

Example

Real-time multimedia benefits from an edge server near a base station, while large archival analytics may use the central cloud.

Conclusion

Edge computing improves latency, jitter and efficiency, whereas MCC provides greater centralized resource capacity.

VERY IMPORTANT EXAM SUGGESTION: Use a six-row table: deployment, distance, latency, jitter, computational power and storage capacity.

Rapid Revision Tables

1. Service Models at a Glance

Model	Provider supplies	User controls	Lecture example
SaaS	Complete application and full stack	Use/configuration	Office 365, Google Apps
PaaS	Runtime, middleware, OS and infrastructure	Applications and data	Azure, Google App Engine
IaaS	Networking, storage, servers and virtualization	OS, middleware, data and applications	Amazon EC2

2. Deployment Models at a Glance

Model	Access/ownership	Main strength	Main concern
Public	Provider-owned, many customers	Low setup cost and scalability	Lower dedicated control
Private	One organization	Control and security	Maintenance
Hybrid	Combination of distinct clouds	Portability and flexibility	Integration
Community	Organizations with common concerns	Shared compliance needs	Joint governance
Multi-cloud	Multiple provider services	Fault tolerance and choice	Heterogeneity

3. Fog vs MEC vs Cloudlet

Aspect	Fog	MEC	Cloudlet
Typical node	Router/gateway/AP	Server at base station	Data center in a box
Location	Between device and cloud	Cellular edge	Near mobile user
Architecture	Fog abstraction layer	Mobile orchestrator	Cloudlet agent
Context awareness	Medium	High	Low
Main access	Bluetooth/Wi-Fi/mobile	Mobile network	Wi-Fi

4. MCC vs Edge Computing

Aspect	MCC	Edge Computing
Deployment	Centralized	Distributed
Distance to user	Far	Close
Latency	Long	Short
Jitter	High	Low
Compute/storage	Abundant	Limited
Network/battery use	Higher	Lower

Final Result Booster: Revise definitions first, then diagrams and comparison tables. In the exam, answer exactly what is asked and use the same terminology as the lecture slides.